Precision aerial photogrammetry system with its own RTK base

Design, construction and in-flight validation of a UAV system with centimetre-level GNSS positioning

Arnau Mariegas

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Summary

This document describes the design, construction and commissioning of a precision aerial photogrammetry system made of three subsystems: an RTK GNSS base of its own, a UAV with two GNSS receivers in *moving baseline* and a camera with hot-shoe timing, and a ground station that links them. The goal is the centimetre-level geotagging of aerial photographs to produce orthomosaics and digital surface models, with a hardware cost of about €2,150 and full control over the firmware, the corrections and the data flow.

The main results are:

- An architecture that serves RTCM3 corrections over local WiFi and forwards them to the drone over a long-range telemetry link, and that obtains RTK Fixed for 100 % of the flight on both receivers, with GNSS-derived yaw and no magnetic compass.
- A flight-test campaign that, from log analysis, resolved control oscillations, a moving-baseline configuration failure, the collapse of a battery and the camera triggering problems (chapter 3).
- Two photogrammetry flights processed end to end: 118 photographs reconstructed at 30 m, with a GSD of 0.45 cm/px and a reprojection error of 0.90 px, and 110 at 50 m (chapter 4).
- The measurement, by two independent methods, of an effective shutter delay of 0.27–0.28 s (∼1.3 m at 5 m/s), which justifies the hot-shoe feedback and is compensated in the geotagging.
- An analysis of the propulsion group that identifies the propeller diameter as the limiting factor for endurance and quantifies the room for improvement (section 3.6).

The accuracy of the final product cannot yet be stated: it requires ground control and check points measured with the same base as the flight, planned in the precision test plan (section 2.11). Because the base runs a *survey-in* in every session, the absolute position of the map has an offset of the order of metres that can only be corrected with the base log. The document marks with a *Pending* box the documentation and tests that are still missing, and the consolidated task list is in section 3.7.

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Part I

Design and construction

Chapter 1

System design and construction

1.1 Introduction

This document describes the design, construction and in-flight validation of a complete precision aerial photogrammetry system (drone *surveying*): an RTK GNSS base of its own, a UAV with two GNSS receivers and a camera with hot-shoe timing, and a ground station that links them.

The goal is centimetre-level geotagging of aerial photographs ($\leq$ 2 cm horizontal, $\leq$ 3 cm vertical) to produce survey-grade orthomosaics and digital surface models, with a total hardware cost of around €2,150 – a fraction of the cost of a DJI P4 RTK or of an Emlid + UAV set – and with full control over the *firmware*, the corrections and the data flow.

The system is divided into three subsystems:

- **RTK GNSS base**: fixed station that generates RTCM3 corrections.
- **UAV (*rover*)**: drone with two GNSS receivers for RTK positioning and *moving baseline* heading.
- **Ground station (GS)**: telemetry link and relay of the corrections to the drone.

The document is organised in four blocks: design and construction (this chapter), the operation manual and procedures (chapter 2), the flight-test and tuning campaign (chapter 3) and the photogrammetry flights that result from it (chapter 4).

1.2 Architecture and data flow

The chosen architecture uses the base to serve RTCM3 corrections over local WiFi (*access point* mode) through a TCP server of its own, without NTRIP. A direct base $\rightarrow$ drone radio link was discarded to simplify the system (a single radio channel to the drone) and to gain resilience in rural areas without mobile coverage.

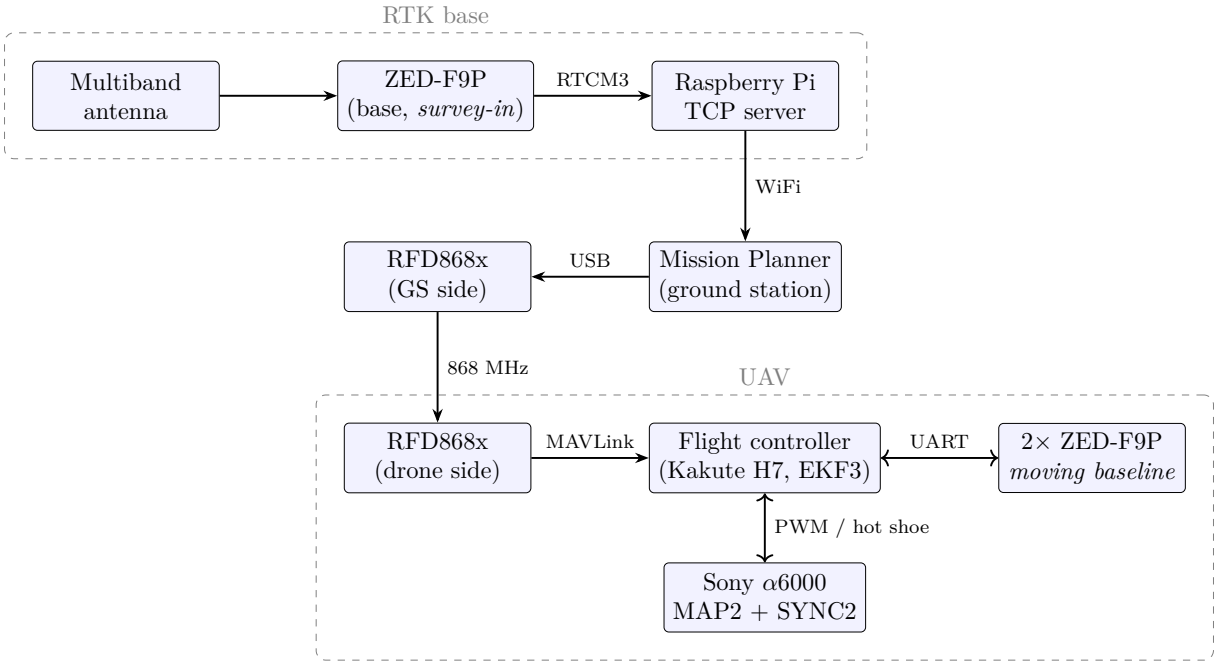


Figure 1.1: Block diagram of the system and flow of the RTCM3 corrections.

1.2.1 Complete data flow

1. The multiband antenna receives L1/L2/L5 (GPS, GLONASS, Galileo, BeiDou).
2. The base **ZED-F9P** (*survey-in* mode in every session, section 2.8) generates **RTCM3** messages (1005, 1074, 1084, 1094, 1124 and 1230) at 1 Hz. Message 1005, with the base position, is only emitted when the *survey-in* is valid.
3. The **Raspberry Pi** receives these messages over USB (serial port of the simpleRTK2B Pro receiver) and redistributes them as they arrive through the TCP server on port 5000, over WiFi AP.
4. The **GS** (Mission Planner / QGroundControl at the pilot's side) connects to the TCP server and forwards the RTCM3 stream to the drone over MAVLink (`GPS_RTCM_DATA`) through the **RFD868x** pair.
5. On the drone, the primary F9P receives the corrections, computes the RTK position and, in *moving baseline* mode, generates RTCM 4072.x for the secondary to derive the *heading*.
6. The **FC** integrates the RTK position and the GNSS *yaw* in the EKF3 for navigation and geotagging.

1.3 Design decisions

Moving baseline with 2× ZED-F9P instead of a UM982 with native *heading*. The F9P is better documented, has more community support and is interchangeable between base and UAV. It provides GPS-for-yaw, which makes it possible to do without the magnetic compass – an important advantage on *frames* with a lot of current.

RFD868x for the GS↔UAV link at 868 MHz (European band) it allows up to 500 mW EIRP and a range above 40 km. Always the European 868 version (never 915 or 433).

Local TCP server over WiFi robust in the absence of mobile coverage; in practice it is almost always operated locally (park the vehicle, set up the tripod next to it and fly).

Hot-shoe feedback (Seagull SYNC2) solves the variable shutter latency (∼100 ms equals

1.5 m of error at 15 m/s). Without this *feedback* there is no real centimetre-level precision in the geotagging; section 4.4 quantifies the effect.

Optional PPK post-processing logging of raw `UBX` data at the base and on the drone, which gives a solution independent of the quality of the real-time RTK link.

1.4 Bill of materials

The hardware of the system has a total cost of €2,150.

1.4.1 Drone

Table 1.1: Drone components (UAV/*rover*)

Component	Model	Approx. cost
Frame	S500 (Quad X, 500 mm between axes)	€35
Motors (4×)	Pilotix 2808, 1100 KV	€60
ESC	Hobbywing XRotor 65A 4-in-1 (DShot)	€30
FC	Holybro Kakute H7 (STM32H7)	€55
GPS (2×)	Holybro H-RTK F9P Helical	€510 (€255 each)
Telemetry	RFD868x EU 500 mW (drone + GS pair)	€363
RC	Radiomaster XR1 Nano ELRS dual-band	€16
Battery	Gaoneng GNB 6S 4000 mAh 70C	€70

Pilotix 2808 motors, 1100 KV moderate KV for 6S, intended for 8-inch propellers (table 3.11) with the additional payload of the RTK + camera system, instead of the high KV typical of *freestyle*, which sacrifice efficiency for speed. The 7-inch propellers of the initial build were used for availability, not by design (section 3.6.1).

Hobbywing XRotor 65A 4-in-1 ESC (DShot) current margin for the additional weight; bidirectional DShot gives per-motor RPM telemetry, useful for vibration diagnosis; the 4-in-1 format simplifies the power wiring.

Holybro Kakute H7 FC (STM32H7) enough UART ports for the whole architecture without any expansion (2×F9P, RFD868x, ELRS RC, ESC telemetry, USB) and native ArduPilot support for *moving baseline* GPS and DShot.

Holybro H-RTK F9P Helical GPS (2×) integrated module (helical antenna + F9P *breakout*) already designed for *moving baseline* with ArduPilot; avoids having to design and match antenna and *breakout* separately.

RFD868x telemetry full justification in section 1.3.

Radiomaster XR1 Nano ELRS RC minimum size and weight so as not to penalise the payload; low-latency ELRS complements the RFD868x, which carries the MAVLink/RTCM data but not the flight control.

Gaoneng 6S 4000 mAh 70C battery 6S for better voltage efficiency with the additional consumption (GPS, telemetry, camera); 70C gives enough peak-current margin with the 4 motors at full load.

1.4.2 GNSS base

Table 1.2: RTK base components

Component	Model	Approx. cost
GNSS	ArduSimple simpleRTK2B Pro (ZED-F9P)	€126
Antenna	BD900 GNSS (multiband)	€43
SBC	Raspberry Pi Zero W	€35
Power	Waveshare UPS HAT + 2× 18650 3500 mAh	€47

simpleRTK2B Pro GNSS (ZED-F9P) development board with USB and an SMA antenna connector, with the same chip as the drone receivers, which gives interchangeability and a shared configuration and *firmware* (section 1.3).

BD900 GNSS antenna (multiband) necessary to receive the two bands and the multiple constellations used by the ZED-F9P, a requirement for the *survey-in* to converge (section 2.8).

Raspberry Pi Zero W small size and low consumption for field use on battery, with enough capacity to distribute the corrections through the TCP server and act as a WiFi access point (NetworkManager).

UPS HAT + 2×18650 power base endurance independent of the mains in the field, with protection against cuts during logging.

1.4.3 Ground station

Table 1.3: Ground station (GS) components

Component	Model	Approx. cost
Transmitter	Radiomaster TX16S (or similar) ELRS	€167
Telemetry	RFD868x (GS side, included in the pair)	–
Laptop / tablet	Own, with Mission Planner	–

The transmitter and the RFD868x pair are standard items with no specific design decision. The only additional element of the GS is the laptop or tablet with Mission Planner or QGroundControl, which bridges the base TCP server and the MAVLink stream to the drone (section 1.2).

1.4.4 Camera (mapping module)

Table 1.4: Camera module for photogrammetry

Component	Model	Approx. cost
Body	Sony α6000 (24 MP APS-C, second hand)	€200
Lens	TTArtisan 25 mm f/2 APS-C, manual focus	€71
Trigger	Seagull MAP2 (fires by PWM)	€30
Feedback	Seagull SYNC2 (hot shoe, X-sync → FC)	€30
SD card	3× SanDisk Extreme 64 GB	€45

Sony α6000 body large APS-C sensor (good signal-to-noise ratio) with 24 MP resolution, in a reasonable body size and weight for the drone; second hand to keep the cost down.

TTArtisan 25 mm f/2 short focal length for a wide field of view (more coverage per flight pass and fewer passes needed) and enough speed for short exposures. Manual focus allows fixing the focus at infinity and prevents the autofocus from hunting during flight, which matters for photogrammetry. Compact, light metal construction, at a much lower cost than a native lens.

***Trigger* Seagull MAP2** remote PWM trigger from the FC without having to modify the camera electronically.

***Feedback* Seagull SYNC2** mandatory for centimetre-level geotagging precision; justification in section 1.3.

1.4.5 Extras and consumables

JST-GH cables, XT60/XT90 connectors, M3 *standoffs*, carbon *bracket* for the helical antennas, 5/8-inch-thread survey tripod, LMR-240 coaxial cable, TNC/SMA connectors with IP54 protection, microSD cards and tools (soldering iron, JST-GH crimper, multimeter), at a total cost of about €218.

1.5 Assembly

1.5.1 Drone (UAV)

The drone is built on an S500 *frame* in X configuration, 500 mm between axes. A side carbon plate is mounted on the central board and acts as the *baseline bracket*: at each end there is a vertical carbon post carrying a Holybro H-RTK F9P Helical module, so that the two antennas are parallel, pointing up, at the highest point of the aircraft (fig. 1.2). The separation between antennas is 0.38 m along the lateral axis of the drone (section 2.1). The RFD868x telemetry module, with its two antennas, is fixed on the top plate and connected to the flight controller by UART.

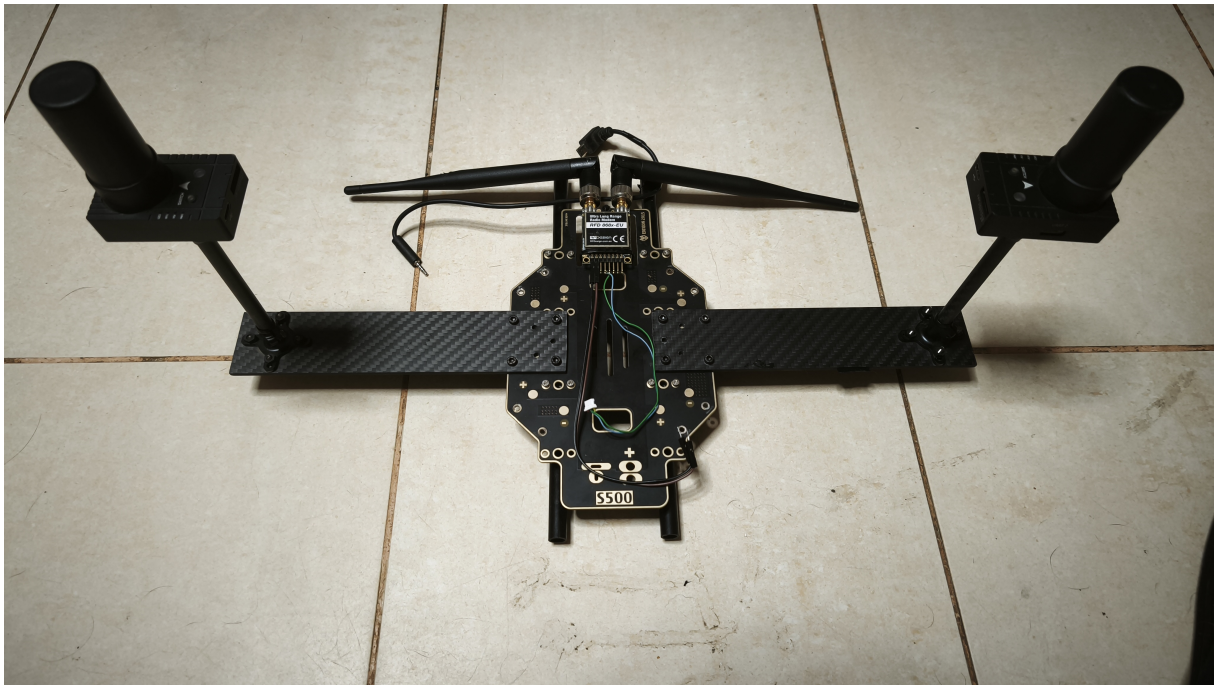


Figure 1.2: S500 central board with the carbon *bracket*, the two H-RTK F9P Helical modules on vertical posts and the RFD868x module.



Figure 1.3: Finished drone seen from above: motors, lateral *baseline* with the two GNSS antennas, telemetry module and landing gear.

Under the *frame* there is a rail made of two parallel carbon bars with clamps (fig. 1.4), which supports both the camera and the battery. The camera (Sony α6000 with the 25 mm lens) points at nadir, with the Seagull SYNC2 on the hot shoe; the 6S battery is fixed to the same rail, so that the centre of gravity is close to the vertical axis of the drone.



(a) Lower rail without load.



(b) With the camera, the SYNC2 and the battery.

Figure 1.4: Bottom view of the *frame* (turned over): carbon rail and camera module.

The configuration of the FC serial ports and the *firmware* parameters (camera, *moving baseline* GPS) are described in the operation manual (chapter 2).

Electrical connections

Figure 1.5 shows the wiring of the drone and table 1.5 summarises it pad by pad. All devices connect directly to pads of the flight controller; no GNSS receiver shares a cable with the other (the *moving baseline* stream goes through the controller, section 2.4).

Table 1.5: Flight controller connections (Kakute H7 V1.2)

Device	FC pads	Function	ArduPilot port
GPS1 (MB-Base)	R3, T3, 5V, GND	UART	`SERIAL3`
GPS2 (MB-Rover)	R4, T4, 5V, GND	UART	`SERIAL4`
ELRS receiver	R6, T6, 5V, GND	UART (CRSF)	`SERIAL6`
RFD868x	R1, T1 (JST-GHR-6P); 5V, GND	UART (MAVLink)	`SERIAL1`
4-in-1 ESC	JST-SH-8P: B+, CUR, R7, GND, M1–M4	DShot, voltage, current, telemetry	`SERIAL7`
Seagull MAP2	M5; 5V, GND	*trigger* PWM	`SERVO5`
Seagull SYNC2	M6; 5V, GND	hot-shoe *feedback*	`SERVO6` (GPIO 55)

1.5.2 GNSS base

The base is a compact stack of three boards fixed to one leg of the tripod with clamps (fig. 1.7): the ArduSimple simpleRTK2B Pro receiver (with the ZED-F9P), the Raspberry Pi Zero W and the Waveshare UPS HAT with the two 18650 batteries. The BD900 GNSS antenna is mounted levelled, with a ground plane, on the 5/8-inch thread of the survey tripod, at a height of 122 cm in the flight tests.

The connections are (fig. 1.6):

- **Antenna → simpleRTK2B Pro:** coaxial cable with SMA connector.
- **simpleRTK2B Pro → Raspberry Pi:** one USB-C cable, which carries the data (RTCM3 corrections and receiver configuration) and powers the receiver from the Pi.
- **UPS HAT → Raspberry Pi:** only the power pins of the Pi header are connected to the UPS HAT, which powers it from the batteries; there is no communication line between the two.

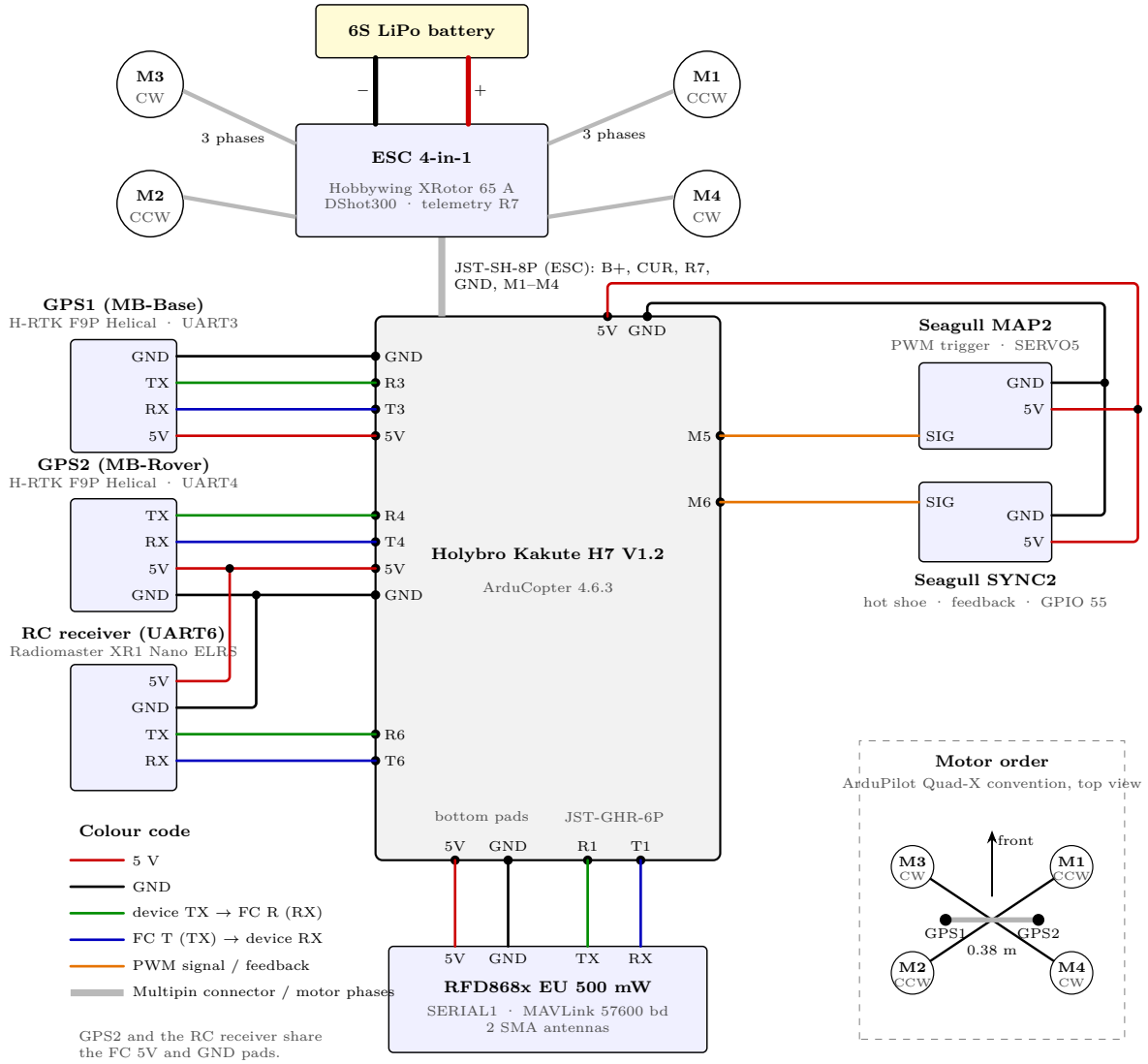


Figure 1.5: Connection diagram of the drone. The colours follow the legend: green for the TX line of the device towards the RX of the FC and blue for the TX of the FC towards the RX of the device.

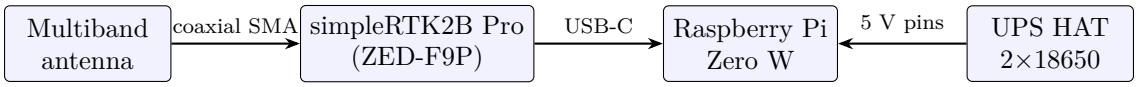


Figure 1.6: RTK base connections: antenna, receiver, SBC and power.



(a) Complete base on the tripod.



(b) Base stack fixed to a leg.

Figure 1.7: RTK base: multiband antenna on the tripod and stack with the simpleRTK2B Pro, the Raspberry Pi and the UPS HAT.

The procedure for positioning the base and fixing its coordinates is described in section 2.8.

Software and configuration. The Raspberry Pi runs Raspberry Pi OS (Debian 13, 32-bit). The corrections are distributed by the RTKLIB `str2str` program (the `rtklibexplorer` fork, compiled on the Pi itself), run as a `systemd` service that starts with the board. The service is called `rtkbase`, but it has no relation to the `rtkbase` project in the bibliography: it reads the USB serial port of the receiver (115,200 bd) and opens a TCP server on port 5000 that forwards the receiver stream as it comes out (RTCM3 and the UBX navigation messages that it emits on the same port), without NTRIP. Besides forwarding it, the service saves it to a file for the whole session (section 2.8). The GS connects to it with Mission Planner's *RTK/GPS Inject* in *TCP Client* mode, at `192.168.4.1:5000`.

The WiFi access point (`BaseRTK`) is managed by NetworkManager in *shared* mode: 2.4 GHz band, channel 6, WPA-PSK, address `192.168.4.1/24` with DHCP, and it is activated at every boot. PMF (802.11w) must be disabled, because the `brcmfmac` driver of the Pi does not support it in AP mode.

The ZED-F9P (*firmware* HPG 1.51, protocol 27.50) is configured with `ubxtool`: NMEA disabled on the USB port, RTCM3 enabled and messages 1005, 1074, 1084, 1094, 1124 and 1230 at 1 Hz. The total stream on port 5000, with the UBX navigation messages that the receiver also emits, was about 15 kbps. The base position is not fixed: the receiver runs a *survey-in* in every session, with 300 s as a minimum and until the precision of the mean is 2 m or better, and the configuration is kept in the *flash* memory of the receiver. Flight tests 5 and 6 used this base and this method, with the antenna at 122 cm.

Limitation. In flight tests 5 and 6 the service only forwarded the stream, without saving it, and the position resulting from the *survey-in* was not noted down; therefore the absolute position of those flights cannot be recovered. Since every session performs a new one, the absolute position of the map has a common error of the order of metres; this shifts the whole map but does not change its relative geometry. Since 27 October 2026 the service saves the stream of every session, so the position can be computed afterwards (section 2.8).

1.6 System status

The hardware is complete and the system has accumulated about 30 minutes of flight according to the controller counter, spread between the tuning campaign (chapter 3) and two photogrammetry flights (chapter 4). With the current configuration the drone flies with RTK Fixed on both receivers and valid GNSS *yaw*, and the photographic chain (distance-based triggering, hot-shoe *feedback*, geotagging and processing) is operational end to end. The current limitations – above all the endurance and the absolute validation with control points – are collected in sections 3.5 and 4.5, and the pending work is in section 3.7.

Chapter 2

Operation manual and procedures

This chapter collects the reference configuration of the system and the working procedure, applicable to any photogrammetry flight. The results obtained with these procedures are presented in chapter 4.

2.1 GNSS antennas

2.1.1 Drone helical antennas

The two helical antennas of the *rover* are mounted:

- Vertical, parallel to each other, pointing up.
- At the highest point of the drone.
- With a minimum separation of 10 cm from the VTx, the ESCs and the power cables, to avoid interference.
- Without tilt with respect to the vertical axis.

2.1.2 *Moving baseline* baseline

The separation between the two helical antennas is **0.38 m**, on a rigid carbon *bracket* aligned with the Y (lateral) axis of the drone. The *offsets* with respect to the centre of gravity are $Y = \mp 0.19$ m and $Z = -0.10$ m (GPS1 on the left, GPS2 on the right), with $X = 0$ (table 2.4).

The *heading* precision of a *moving baseline* scales inversely with the baseline length: with 0.38 m a σ of the order of 0.5° is estimated, which must be validated empirically (section 2.11). Any change in the length of the *bracket* requires recalibrating the GNSS *heading*.

2.1.3 Base antenna

Horizontal *patch* antenna with a ground plane, levelled on the survey tripod.

2.1.4 Camera timing

The central pin of the camera *hot shoe* is the **X-sync** signal: the Seagull SYNC2 fits directly onto it, since the Sony α6000 has a *Multi Interface Shoe* compatible with the ISO 518 standard.

2.2 FC serial port configuration (Kakute H7)

Table 2.1: Serial port assignment of the Kakute H7

Port	Function	Baud rate	Protocol
SERIAL0 (USB)	MAVLink2 (configuration)	–	–
SERIAL1 (UART1, VTX port)	RFD868x MAVLink	57600	2
SERIAL2 (UART2)	Internal Bluetooth (occupied)	–	Disabled
SERIAL3 (UART3)	F9P #1 – GPS1 (MB-Base)	230400	5
SERIAL4 (UART4)	F9P #2 – GPS2 (MB-Rover)	230400	5
SERIAL6 (UART6)	RC ELRS (CRSF)	420000	23
SERIAL7 (UART7, ESC port)	ESC telemetry	115200	16

2.3 Camera configuration

2.3.1 Controller parameters

From ArduPilot 4.5 on, the camera parameters have the prefix `CAM1_*` to support multiple cameras; the old documentation and forums still use `CAM_*` names.

Table 2.2: Camera parameters (ArduPilot 4.5 or later)

Parameter	Value / Function
`SERVO5_FUNCTION`	10 (*CameraTrigger*, MAP2 on pad M5)
`CAM1_TYPE`	1 (Servo)
`CAM1_DURATION`	1.0 (in **seconds**, not ms): the MAP2 in IS-T mode needs commands of 1 s
`CAM1_SERVO_ON`	1700 (IS-T mode)
`CAM1_SERVO_OFF`	1500 (inside the MAP2 rest range, 1400–1600)
`SERVO5_TRIM`	1500
`CAM1_TRIGG_DIST`	0 in the base configuration; set for each mission (section 2.7)
`SERVO6_FUNCTION`	−1 (GPIO for the SYNC2 *feedback*, pad M6)
`CAM1_FEEDBAK_PIN`	55
`CAM1_FEEDBAK_POL`	1

The parameter name is `CAM1_FEEDBAK_PIN`, spelled **without the "C"** of *feedback* (historical ArduPilot naming). Pin 55 corresponds to the `SERVO6` pad reassigned as GPIO.

With a manual lens without electronic contacts, the MAP2 must not work in AF-T mode (useless 1 s pre-focus, which adds a constant delay). The diagnosis is detailed in section 3.2.6.

2.3.2 Camera settings

Table 2.3: Sony α6000 settings for photogrammetry

Setting	Value
Exposure mode	M (manual)
Shutter speed	1/1600 s
Aperture	f/5.6 (ring fixed with tape)
Focus	Manual at infinity (both rings fixed with tape)
ISO	Auto, with a ceiling at 1600
Drive mode	*Single shot*, JPEG Fine
Shutter	Electronic front curtain enabled
Release w/o Lens	Enabled (needed with a lens without contacts)

The manual lens does not send data to the camera: the EXIF records `FocalLength=0` and an empty `LensModel`, and the orientation would depend on the heading (section 4.1.2). This metadata is rewritten in post-processing.

The camera has no dynamic range to compensate for low light: photographs must be taken with the sun more than 10–15° above the horizon (section 4.3).

2.3.3 Shutter delay

The controller trigger command (`TRIG`) and the actual exposure are separated by ≈ 0.28 s (section 4.4); at 5 m/s that is ≈ 1.4 m. For this reason the geotagging uses the position of the `CAM` message (instant of the hot-shoe *feedback*) when it is available, and compensates with $v \cdot \tau$ when it is not.

2.4 *Moving baseline* GPS and *yaw* configuration

Table 2.4: *Moving baseline* and GPS-for-yaw parameters

Parameter	Value
`GPS1_TYPE`	17 (MB-Base)
`GPS2_TYPE`	18 (MB-Rover)
`GPS_AUTO_CONFIG`	1
`GPS_DRV_OPTIONS`	0
`GPS_INJECT_TO`	127
`EK3_SRC1_POSXY` / `VELXY`	3 (GPS)
`EK3_SRC1_YAW`	2 (GPS yaw)
`COMPASS_ENABLE`	0 (no compass)
`GPS1_POS_X/Y/Z`	0 / −0.19 / −0.10 m
`GPS2_POS_X/Y/Z`	0 / +0.19 / −0.10 m
`GPS1_MB_OFS_Y`	−0.38 m

`GPS_DRV_OPTIONS` must be 0: the value 1 (use of UART2 for the *moving baseline*) assumes a direct cable between the UART2 ports of the two F9Ps to carry the RTCM stream, which does

not exist in this build; with the bit set, the *rover* never reaches RTK Fixed (`PreArm: GPS[2] yaw not available`). The stream between receivers goes through the flight controller.

2.5 Control, battery and safety

Table 2.5: Control, battery and safety parameters of the reference configuration

Parameter	Value
FRAME_CLASS	1 (Quad)
FRAME_TYPE	1 (X)
MOT_PWM_TYPE	5 (DShot300) [1]
ATC_RAT_RLL_P / ATC_RAT_PIT_P	0.07
ATC_RAT_RLL_I / ATC_RAT_PIT_I	0.07
ATC_RAT_RLL_D / ATC_RAT_PIT_D	0.002
ATC_RAT_YAW_P	0.10
ATC_RAT_YAW_I	0.01
MOT_THST_EXPO	0.65
MOT_SPIN_ARM	0.04
MOT_SPIN_MIN	0.06
MOT_THST_HOVER	0.225 (learned in flight, MOT_HOVER_LEARN=2)
BATT_MONITOR	4 (analogue voltage and current)
BATT_VOLT_MULT	10.82 (calibrated)
BATT_AMP_PERVLT	87.73 (calibrated)
BATT_LOW_VOLT / BATT_FS_LOW_ACT	21.6 V / 2 (RTL)
BATT_CRT_VOLT / BATT_FS_CRT_ACT	20.4 V / 1 (Land)
BATT_FS_VOLTSRC	0 (instantaneous voltage)
FENCE_ENABLE / FENCE_TYPE	1 / 7 (altitude, circle and polygon)
FENCE_RADIUS	700 m
FENCE_ALT_MAX	120 m
WPNAV_SPEED	500 cm/s (5 m/s) in automatic missions
FLTMODE1 to FLTMODE6	Stabilize, AltHold, Loiter, Auto, Guided, RTL
LOG_BITMASK	196607

2.6 Pre-flight checklist

1. Verify that both F9Ps (base and drone) report `RTK Fixed` before arming, with the correct *fix* type on both receivers (not `RTK Float` or `3D Fix`).
2. Verify that the GNSS *yaw* (*moving baseline*) is valid and consistent with the real *heading* (margin $< 1°$).
3. Check the RFD868x GS↔drone link and the reception of RTCM corrections on the drone.
4. Check the battery level of the drone (voltage per cell) and of the base (UPS).
5. Check the SD card and the camera battery, and that the settings are those of table 2.3.
6. Test the trigger with *Trigger Camera Now* and confirm that `CAM` messages appear (SYNC2 *feedback* active).
7. Review the assigned flight modes and that the *geofence* is active (table 2.5).
8. Check in Mission Planner that the EKF3 is healthy (no position or velocity variance warnings), that the HDOP is low (recommended < 1 with RTK Fixed) and that there is no pending *prearm* message.

9. Verify that `COMPASS_ENABLE=0` and that there is no compass *prearm error*.
10. Check the *failsafes*: loss of RC, loss of telemetry, low and critical battery.
11. Check that the sun is more than 10–15° above the horizon and that the flight area and the airspace allow the flight (section 2.14).

2.7 Flight planning

2.7.1 GSD calculation

The *Ground Sample Distance* depends only on the flight height H, the sensor size, the focal length f and the image resolution:

$$\mathrm{GSD}\,(\mathrm{cm/px}) = \frac{H\,(\mathrm{m}) \times \text{sensor width}\ (\mathrm{mm}) \times 100}{f\,(\mathrm{mm}) \times \text{image width}\ (\mathrm{px})}$$

With the Sony α6000 (APS-C sensor of 23.5 × 15.6 mm, 6000 × 4000 px, pixel size ≈ 3.92 µm) and the 25 mm TTArtisan lens:

$$\mathrm{GSD}\,(\mathrm{cm/px}) \approx 0.0157 \times H\,(\mathrm{m})$$

Table 2.6: GSD and coverage per image as a function of height (Sony α6000 + TTArtisan 25 mm)

Height (m)	GSD (cm/px)	Footprint (m, width × height)
30	0.47	28.2 × 18.7
50	0.78	47.0 × 31.2
60	0.94	56.4 × 37.4
80	1.25	75.2 × 49.9
100	1.57	94.0 × 62.4
120	1.88	112.8 × 74.9

The values measured in the flight tests agree with this relation: taking into account the focal length calibrated by the processing (25.72 mm), a GSD of 0.451 cm/px corresponds to a height of 29.6 m (flight planned at 30 m), and a GSD of 0.743 cm/px to 48.9 m (flight planned at ∼50 m).

2.7.2 Flight parameters

Table 2.7: Adopted and recommended flight parameters

Parameter	Value
Flight height	According to the target GSD (table 2.6); 30 m and 50 m in the flight tests
Front / side overlap	75–77 % / 65–69 % in flight 5; 80 % / 70 % recommended with complex relief
Speed	5 m/s (`WPNAV_SPEED=500`); limited by *motion blur*: $v_{\max} = \mathrm{GSD}/t_{\mathrm{exposure}}$
Triggering	By distance (`CAM1_TRIGG_DIST`), with the interval between photos greater than the 1 s of `CAM1_DURATION`
Camera orientation	Long side of the sensor along the flight direction
Lighting	Sun more than 10–15° above the horizon

The 80/70 % overlap percentages are the usual default value in photogrammetry software for precision projects with well-textured terrain; they can be reduced with caution thanks to RTK geotagging, but insufficient overlap still limits the quality of the SfM *matching*. Since `CAM1_DURATION=1.0` sets a minimum interval of about 1 s between triggers, at 5 m/s the distance between photos must be at least 5 m (section 3.5.1).

2.8 Field protocol

2.8.1 Positioning the base

a) Tripod on a stable point with a clear 360° sky, away from walls, metal roofs and vehicles; measure the antenna height (122 cm in the flight tests).

b) Switch on the Pi and wait about 30 s for the `BaseRTK` network to appear; connect the GS to it.

c) Restart the *survey-in* to erase the mean of the previous session and wait until it is valid, from the PC connected to `BaseRTK` and inside the `base_rtk` folder: `python base_rtk.py survey`. The command shows the duration, the observations and the mean error as it goes, and ends when the *survey-in* is valid (minimum 300 s and mean error, `meanAcc`, of 2 m or better) or after 20 min (`-minuts N` changes the limit). While it lasts, the `rtkbase` service is stopped and port 5000 does not serve; it restarts by itself at the end, also if the SSH connection drops or Ctrl-C is pressed.

d) If it does not become valid, the mean stalls because of *multipath* or sky obstruction: improve the antenna location, do not extend the wait. The receiver keeps the *survey-in* running, and `python base_rtk.py survey -estat` queries it without restarting it.

e) When the *survey-in* is valid, the command automatically saves the ECEF position (`meanX/Y/Z`) and the final `meanAcc` to `survey.txt`, inside the active session on the Pi. It is the only record of the base position for that session, and it is downloaded with the log (see "Session log" below).

While the *survey-in* is not valid, the stream does not carry message 1005 (base position) and the drone cannot reach RTK Fixed. The base position is therefore that of the *survey-in*, with an error of the order of metres that shifts the whole session (section 2.11.1). To state absolute positions it will be necessary to reuse a fixed position recorded at the same point, or to correct the offset of each session with a PPP solution from its observation log.

Session log. The `rtkbase` service saves the receiver stream to files while forwarding it on port 5000, for the whole session: `str2str` has two outputs, `tcpsvr://:5000` and `file:///home/arnau/logs/actual/base_%Y%m%d_%h%M.rtcm3::S=1`. Each start of the service opens a new *session*, a folder `/home/arnau/logs/sNNNN/` (`actual` points to the one in progress), and a file is created inside it every hour (`::S=1`). The file names carry the UTC time of the Pi, and the date is unreliable, because the Pi has no real-time clock or internet: the valid time is the GNSS time inside the data. A service restart or a power cut opens a new session, except for what `survey` does, which continues in the same one. The log is the same stream as the port, about 7 MB per hour, with the RTCM3 observations (pseudorange, phase and C/N0 of the four systems), message 1005 once the *survey-in* is valid and the UBX navigation messages that the receiver emits on the same port. It carries no ephemerides, which must be taken from a nearby reference station, nor Doppler.

At the end of the session the log is downloaded before switching off the base, from the PC connected to `BaseRTK`, with `python base_rtk.py baixa` (the last session) or `python base_rtk.py baixa N` (session `N`; `python base_rtk.py llista` shows the existing ones, with size and hours). The command joins the hourly files of the session into a single one, `base_sNNNN.rtcm3`, checks

its size and, if there are *survey-in* notes, also downloads `base_sNNNN_survey.txt`. `-o folder` says where they are left, and nothing is deleted from the Pi. The file can be converted to RINEX with `convbin -r rtcm3` to compute afterwards the session position with a PPP solution or relative to a reference station. This computation has not yet been done with any log.

The Pi disk holds about 3,500 h of log and there is no automatic deletion: sessions that are no longer needed are deleted by hand with `python base_rtk.py esborra N` or `esborra -abans N`. The command asks for confirmation and never deletes the active session. The tools and their installation are in the `base_rtk` folder, with its `README.md`.

2.8.2 Verifying the *moving baseline*

a) Measure the real vector between the two antennas.

b) Check the GNSS *yaw* against the real *heading* (margin $< 1°$).

2.8.3 Flight

a) Base in `Fixed` → the GS injects RTCM into the drone.

b) Arm **only** with RTK Fixed on both F9Ps and valid *yaw*.

c) Fly the planned *grid* in Auto mode.

d) Download the FC *logs* (`.bin`) and the photographs from the SD card.

2.8.4 Post-processing

a) Pair each photograph with its `CAM` (or `TRIG`) message from the *log* and rewrite the EXIF (section 4.1.2).

b) Generate the geotag file with the per-photo precision.

c) Process with ODX/WebODM (section 2.10) and generate orthomosaic, DSM and point cloud.

2.9 Ground control points (GCP)

The precision of the final product can only be stated with independently measured points. Since the base runs a *survey-in* in every session, the **coordinate frame** of the points is as important as their precision. The planned protocol is:

- Targets well distributed over the whole area, including the corners and the centre, marked with a high-contrast pattern recognisable in the photograph: at least five control points and, separately, ten check points (section 2.11.2).
- Position of each target measured with RTK against **the same base and the same session** as the flight, with an expected precision of 1–2 cm. This way the points and the *geotags* share the *survey-in* offset and it cancels in the comparison. If they are measured with a total station or another base, a transformation must be fitted before comparing; otherwise the RMSE is dominated by the metre-level base error.
- Separate control points (used in the adjustment) from check points (which do not take part in it), and compute the H/V RMSE of the latter.

Pending. No target has yet been deployed and no control point measured: this is the planned protocol, not a result. It is the task that unlocks absolute precision.

2.10 Processing (*Structure from Motion* and PPK)

Photogrammetric processing is done with the ODX 3.8.3 engine (WebODM 3.3.0, NodeODX 2.3.1) on a PC with an i5-10400F, 32 GB of RAM and an RTX 2060. A flight of 118 photographs takes about 42 minutes. The options that depart from the defaults are: orthophoto resolution of 1 cm/px, DEM resolution of 2 cm/px, high point-cloud quality, DSM generation, *auto-boundary*, COG, glTF and EPT *point cloud*. The detail of each processing run is in chapter 4.

As an alternative or complement to real-time RTK, **PPK** post-processing with RTKLIB is contemplated, from the raw `UBX` recordings of the base and the drone, which gives a positioning solution independent of the quality of the RTK link during the flight.

Pending. PPK has not yet been applied to any flight: the current processing runs use the real-time RTK positions of the controller.

2.11 Quality control and validation criteria

2.11.1 Effect of the *survey-in* on validation

With the *survey-in* the base computes its position as the mean of autonomous positions, without corrections, and broadcasts it in message 1005. The error of this position is transmitted entirely to the *rover* [2]: the whole session is shifted by the same vector, which is of the order of metres (the acceptance criterion of the protocol is a `meanAcc` of ≤ 2 m). In contrast, the *relative* position with respect to the base is barely affected: the baseline-length error is that of the base multiplied by the baseline and divided by the distance to the satellites ($\sim 2 \cdot 10^7$ m), about 0.02 mm with a base shifted by 2 m and a 200 m baseline.

- **Not affected** (can be validated without an absolute reference): the accuracy of the RTK baseline, the repeatability of the *rover*, the *heading* and the internal geometry of the product (shape and scale).
- **Affected:** the absolute position of the map, the comparability between sessions and any comparison with an external reference. Since every session performs a new *survey-in*, two maps of the same place are shifted from each other by the order of metres. The base antenna height (122 cm) enters the elevation with weight 1:1.

There are three ways, not mutually exclusive, to recover the absolute position: reuse a fixed position recorded at the same point; measure a known point with the same base and correct its offset; or compute the session position by PPP from the base log and apply the correction vector to the whole product. The latter is feasible in principle with the RTCM stream already generated, which contains MSM observations (1074, 1084, 1094, 1124) convertible to RINEX with RTKLIB (`convbin`); this must be verified, as well as that the session lasts enough hours for a centimetre-level PPP.

2.11.2 Precision test plan

Table 2.8: Precision test plan with *survey-in* and current status

Test	Procedure	Reference frame	Criterion	Status
T1	Record `NAV-SVIN` (`meanX/Y/Z`, `meanAcc`) in every session and, in samples, compare it with the PPP solution of the same place	—	Informative: documents the offset	**Pending**; not recorded in flights 5 and 6
T2	Static *rover* at three distances (e.g. 10, 50 and 100 m): RTK base→*rover* vector against an independent measurement (total station or distance meter and level)	Relative to the base (independent of the *survey-in*)	< 2 cm H, < 3 cm V	**Pending**
T3	Static repeatability: 10–15 min on a point; standard deviation H/V and percentage of RTK Fixed	Relative	Consistent with 0.01 m + 1 ppm	**Pending**
T4	*Heading*: GNSS *yaw* against a bearing measured with a total station, in several orientations	Independent of the base	$\sigma < 0.5^\circ$	**Pending**; *yaw* valid in all flights
T5	Photogrammetric product: RMSE of the check points, with and without control points, reporting the number of points	Same frame as the *geotags* (same base and session)	< 3 cm H, < 5 cm V	**Pending**; internal consistency: reprojection 0.90 px, model–GPS CE90 0.59 m
T6	Absolute precision: apply the base PPP vector and check against a known geodetic mark	External frame (EPSG:32631)	To be defined (limited by the PPP)	**Pending**; requires the base log

The comparison with a DJI P4 RTK reference *dataset* only makes sense after T6, because it needs an absolutely georeferenced map.

On the number of points: with only five, the uncertainty of the RMSE is large. Edition 2 of the ASPRS standard asks for a minimum of 30 check points to state a product precision, and admits a smaller number in small UAS projects if the report declares it [3]. For that reason a minimum of ten check points is recommended, in addition to the control points, always reporting the number.

The internal-consistency indicators (reprojection error, difference between the model and the GPS positions of the photos) measure the coherence of the processing, **not the precision**: they do not replace T2–T6. Until these are done, no absolute or centimetre-level precision is claimed.

2.12 Export of results

Each processing run delivers, in the EPSG:32631 coordinate system (UTM 31N), the following products: GeoTIFF orthophoto (1 cm/px at 30 m), GeoTIFF DSM (2 cm/px), dense point cloud (LAZ and EPT), textured models (OBJ and GLB) and a PDF report with the processing statistics.

2.13 Drone maintenance

2.13.1 Before each flight

- Visual inspection of the *frame*, the arms and the *standoffs*: cracks or mechanical play.

- Propellers without cracks or deformation, balanced and with the collars well tightened.
- XT60/XT90 and JST-GH connectors without play or signs of heating.
- Helical antennas well fixed, vertical and without damage to the *baseline bracket*.
- Battery condition (swelling, voltage per cell) and enough charge for the flight plan.
- Camera SD card with enough free space and correctly formatted.

2.13.2 Periodic maintenance

- Check the tightening torque of motors and arms every few outings.
- Check the condition of the motor bearings (noise, play on the shaft).
- Update the ArduPilot *firmware* and the F9P receivers only in a controlled way, verifying parameter compatibility before and after.
- Periodic verification of the GNSS *heading* against a known reference, to detect mechanical drift of the *bracket*.
- Cleaning of contacts and connectors, especially after flying in humid or dusty environments.
- LiPo battery maintenance cycles: storage at resting voltage, no discharge below the manufacturer's threshold, and withdrawal of any *pack* that has suffered a *crash*.

2.14 Regulations and safety

This section is informative and does not constitute legal advice. The EASA regulatory framework for self-built drones is in transition; the current status must be verified before each flight campaign [4].

2.14.1 Drone classification

This UAV is **self-built** (*privately built*): it does not come from a ready-to-assemble *kit* sold as a single product, but from individual components. According to EASA, a self-built drone **cannot carry a class C0–C4 mark**, and therefore is limited, within the *Open* category, to subcategory **A3** (far from people and built-up areas), with a maximum weight of 25 kg [4].

Derived practical requirements:

- Registration as a drone operator with the competent aeronautical authority (in Spain, AESA) and visible identification of the registration number on the aircraft.
- A1/A3 pilot certificate (online training and exam) as a minimum; the operator holds it.
- Flight exclusively in subcategory A3: minimum horizontal distance of 150 m from residential, commercial, industrial or recreational areas, and away from people not involved in the operation.
- Check of the airspace and local restrictions (*geo-awareness* zones, NOTAM, proximity to airports) before each flight.
- Third-party liability insurance, recommended (mandatory in many cases depending on weight and use).

Under the transitional rules, self-built drones without a class mark may operate under the inherited regime (by weight) until 1 January 2027. EASA is developing a permanent regime for self-built drones, roughly equivalent to class C4 (subcategory A3, < 25 kg, pilot with an A1/A3 certificate) [4].

2.14.2 Operational safety

- Do not fly with RTK Float or without a valid GNSS *fix* if the mission depends on centimetre-level precision (it also reduces the quality of the EKF3 position estimate).
- Always define a *geofence* and *failsafe* actions for low battery and link loss.
- Maintain visual contact (VLOS) unless the corresponding authorisation for BVLOS is held.
- Basic risk assessment of the flight area: obstacles, power lines, third parties and weather (wind, rain).

2.15 Lines of continuation

- Absolute validation with control and check points (section 2.9).
- Endurance improvement: change of the propulsion group and Li-ion battery (sections 3.5 and 3.6).
- Configuration of the harmonic *notch* filter and Autotune.
- Migration to UM982: a single module with native *heading*, less wiring and less consumption.
- LiDAR integration (Livox Mid-360) for dense vegetation.
- Multispectral camera for precision agriculture.
- Own carrier PCB (F9P + SBC + UPS integrated).
- Publishing corrections to a community NTRIP network (Centipède [5] or rtk2go).

2.16 Reference projects

- `rtkbase` (Stefal) – base software [6].
- SparkFun RTK Express – KiCad design and *firmware* [7].
- `jacobjhansen/RTCM-Pi` – PPP workflow [8].
- Centipède – community NTRIP network in Europe [5].
- ArduPilot – *moving baseline* and GPS-for-yaw [9].

Part II

Flight tests and tuning

Chapter 3

Flight tests and tuning

This chapter collects the flight-test and tuning campaign carried out between August and October 2026, before operating in *surveys* (chapter 4). It records the incidents found in the flight control, the RTK GPS, the camera module and the propulsion group, the causes identified and the changes applied to solve them.

Table 3.1: Flight sessions of the campaign

Date	Session	Result
6 August	Control tuning	Severe oscillation; *crash*
8 August	Control tuning	Two correct flights; *crash* when changing the orientation
10 August	GPS test	Motor failure in flight; *crash*; *moving baseline* error
27 August	Four flights	Four clean flights, RTK Fixed 100 %
4 September	*Grid* mission	Cut by the *geofence*; battery collapse
8 October	Flight test 5	118 photos; orthomosaic and DSM (section 4.2)
13 October	Flight test 6	Two flights at dawn; one usable batch (section 4.3)
20 October	8-inch propellers	Two aborted arms (section 3.6.7)

3.1 Initial controller configuration

The first review of the controller parameter file detected and corrected the following problems:

- `FRAME_CLASS` unassigned (now Quad, X).
- GPS position *offsets* (`GPS1_POS_*` / `GPS2_POS_*`) at zero; measured with respect to the centre of gravity (table 2.4).
- `EK3_SRC1_YAW` pointing to the compass: changed to GPS *yaw*.
- Compass enabled with no sensor connected: disabled.
- `BATT_MONITOR` at 0: changed to analogue voltage and current.
- `MOT_PWM_TYPE` at PWM: changed to DShot300.
- All flight modes at Stabilize: assigned Stabilize, AltHold, Loiter, Auto, Guided and RTL.
- `SERVO9` assigned to *Scripting1* with no real use: disabled.

3.2 Chronology of incidents

3.2.1 6 August – Severe oscillation and *crash*

Wobbling in the arms when raising throttle, with repeatable lateral drift. In the *log*, the *rate* loop changed sign in 65 % of the samples, with errors of up to 301°/s. It ended with `Crash: Disarming: AngErr=51>30`.

Causes identified:

- Drone heavily overpowered (`MOT_THST_HOVER=0.20`, *hover* at 20 %) with default PIDs designed for a *hover* of 35–50 %.
- Motors saturating at the minimum because of a too high `MOT_SPIN_MIN`.
- `MOT_THST_EXPO=0.8`.
- No *notch* filter, with vibrations of 44 m/s^2 and 42 *clips*.
- A `SAVE_TRIM` done in flight had contaminated `AHRS_TRIM_X`.

3.2.2 8 August – *Crash* due to an orientation change

The first two flights of the day went well (oscillation down from 65 % to 3–6 %, zero *clips*). On the third, when setting `AHRS_ORIENTATION` from 4 to 0, the drone flipped in 0.4 s with the controller saturated opposite to the rotation: pure positive feedback.

The real problem was not the orientation but the polarity of channel 2: *roll* responded correctly, only *pitch* was inverted. It was solved by inverting the *elevator* on the Radiomaster.

3.2.3 10 August – *Crash* due to motor failure

Stable *hover* for 10 s, then sudden loss. Motor M3 was commanded to 2000 with no effect; the current dropped from 16 A to 8.9 A, the voltage recovered and the vibrations fell just before the exit: unmistakable signature of a motor stopped in flight.

The same day it was identified that `GPS_DRV_OPTIONS=1` prevented the *moving baseline*: this value assumes a direct cable between the UART2 ports of the two F9Ps, which does not exist in this build. The *rover* never reached RTK Fixed, hence the `PreArm: GPS[2] yaw not available` (section 2.4).

3.2.4 27 August – Four clean flights

First session without incidents. GPS[0] and GPS[1] at 100 % RTK Fixed, GPS *yaw* available in all samples, vibrations of 11–16 m/s^2 and zero *clips* in the four flights.

3.2.5 4 September – Mission cut and *crash* due to battery

Flight 1: the mission was aborted at *waypoint* 25 of 87 by a *geofence breach*. The initial radius of 300 m was insufficient for the new field (farthest *waypoint* at 566 m; 41 of the 87 outside the radius). The photos came out blurred.

Flight 2: battery collapse. From 22.4 V to 8 V in 25 seconds with only 5 A of load. The counter showed 213 mAh consumed from a 4000 *pack*, impossible: the current sensor was badly calibrated. The *sag* went from 0.63 V to 1.79 V in 18 seconds: internal resistance tripled, a cell dying. The *failsafes* acted correctly (RTL at 758 s, Land at 763 s), but there was no energy left to land.

3.2.6 4–13 September – Camera problems

In flight 1: 121 trigger commands, only 33 confirmed exposures, all with a constant delay of 1.16 s. In flight 2: 50 commands, zero photos.

A first diagnosis attributed the EXIF readings `FNumber=0`, `FocalLength=0` and `LensModel` set to `"--"` to a failure of the electrical contacts. In fact it is the normal behaviour of a manual TTArtisan 25 mm f/2, which has no electronic contacts.

Real causes:

- The MAP2 was in AF-T mode (`CAM1_SERVO_ON=1300`), which does a useless 1 s pre-focus with a manual lens: hence the constant delay.
- `CAM1_SERVO_OFF=1100` outside the rest range expected by the MAP2 (1400–1600).
- `CAM1_DURATION=0.1` too short for the MAP2 to read the command (IS-T mode wants 1.0 s).
- *Release w/o Lens* possibly disabled, which blocks the shutter with a lens without contacts.
- Slow shutter and active autofocus in flight 1 (cause of the blurred photos).
- The *grid* configuration assumed 20 mm when the real lens is 25 mm: all the computed overlap was wrong.

Once the changes were applied, flight 2 of test 6 gave 110 trigger commands, 110 *feedback* confirmations and 110 photographs (section 4.3).

3.3 Parameter changes applied

3.3.1 Control (6 August)

Table 3.2: Control tuning after the oscillation of 6 August

Parameter	From	To
`ATC_RAT_RLL_P` / `PIT_P`	0.135	0.070
`ATC_RAT_RLL_I` / `PIT_I`	0.135	0.070
`ATC_RAT_RLL_D` / `PIT_D`	0.0036	0.0020
`ATC_RAT_YAW_P`	0.18	0.100
`MOT_THST_EXPO`	0.8	0.65
`MOT_SPIN_ARM`	0.10	0.04
`MOT_SPIN_MIN`	0.15	0.06
`ATC_RAT_*_FLTD` / `FLTT`	20	10
`AHRS_TRIM_X` / `Y`	contaminated	recalibrated (≈ 0)
`LOG_BITMASK`	180222	196607

3.3.2 GPS

Table 3.3: GPS configuration changes after the incident of 10 August

Parameter	From	To
`GPS_DRV_OPTIONS`	1	0
`GPS1_POS_Y` / `GPS2_POS_Y`	0	−0.19 / +0.19
`GPS1_POS_Z` / `GPS2_POS_Z`	0	−0.10 / −0.10

3.3.3 Safety and battery

Table 3.4: Safety (*failsafes*, *geofence*) and battery changes after 4 September

Parameter	From	To
`BATT_FS_LOW_ACT`	0	2 (RTL)
`BATT_FS_CRT_ACT`	0	1 (Land)
`BATT_LOW_VOLT` / `CRT_VOLT`	–	21.6 / 20.4
`BATT_AMP_PERVLT`	uncalibrated	87.73
`BATT_VOLT_MULT`	uncalibrated	10.82
`FENCE_ENABLE`	0	1
`FENCE_RADIUS`	300	700
`WPNAV_SPEED`	1000	500
`SR1_*` (RFD telemetry)	0	active

3.3.4 Camera

Table 3.5: Camera changes derived from the diagnosis of section 3.2.6

Parameter	Before	Now
`CAM1_SERVO_ON`	1300 (AF-T)	1700 (IS-T)
`CAM1_SERVO_OFF`	1100	1500
`CAM1_DURATION`	0.1	1.0
`SERVO5_TRIM`	1100	1500

The physical camera setup is in table 2.3. The Mission Planner *grid* is planned with the real 25 mm focal length.

3.4 Measured progression

Table 3.6: Progression of the flight metrics over the campaign

Metric	6 Aug	8 Aug	27 Aug	4 Sep	8 Oct	13 Oct
Roll rate sign changes	65 %	3–6 %	8–9 %	stable	–	–
VibeZ max. (m/s^2)	44	40	11–16	14–18	10.8	14.1–15.7
Clips	42	0–29	0	0	0	0
GPS[0] RTK Fixed	–	–	100 %	100 %	100 %	–
GPS[1] RTK Fixed	–	0 %	100 %	100 %	100 %	–
Hover throttle	20 %	20 %	45 %	45 %	–	–

The 8 and 13 October columns correspond to flight tests 5 and 6 (for 6, the range covers both flights). Vibration was computed from the `VIBE` messages of the *logs*, only with the motors running; the median VibeZ is 3.9 m/s^2 in flight 5 and 5.0–5.3 m/s^2 in flight 6. Dashes are metrics that have not been computed for these flights.

3.5 Endurance and performance improvement

Endurance is the biggest limitation of the current platform. *Hover* at 45 % *throttle* (table 3.6) is stable, but the system carries 520 g of APS-C camera with a propulsion group that was not chosen for efficiency (section 3.6): it works, but the price is flight time. This section compares the improvement options by cost and expected gain; the propulsion group is detailed in the next section.

Table 3.7: Options to improve endurance and performance per battery

Option	Cost	Gain	Comment
Fly at 45–50 m	€0	+60 % area per battery	The *grid* must be recomputed with the 25 mm focal length; it also relaxes the trigger rate (section 3.5.1).
Notch + Autotune	€0	5–10 %	See section 3.5.2.
6S2P Li-ion battery	∼€120	×1.7 flight time	Molicel P42A or P45B. Avoid Samsung 50E: 10 A per cell is little for a *hover* of ∼17 A (section 3.5.3).
8-inch propellers (current motors)	∼€15	+12–18 % endurance	Uses the margin of the 1100 KV motors (section 3.6).
400–500 KV motors + 12–13-inch propellers	∼€140	∼ ×1.8–2 flight time	The current 500 mm *frame* already allows it; FC, F9P, radio and camera stay (section 3.5.4).
Li-ion + new motors and propellers	∼€260	30+ min	The real jump to a *survey* platform.

3.5.1 Raising the flight height (zero cost)

The width of the strip photographed in each pass is proportional to the height ($H \cdot 23.5/25$ m with the current sensor and optics, table 2.6). Flying higher covers the same area with fewer passes and therefore less flight time: the gain is directly the ratio of heights. Flight test 5 was flown at 30 m, so going up to 48 m gives +60 % of area per battery (45 m: +50 %; 50 m: +67 %), as is already done in flight test 6 (50 m).

The cost is GSD: at 50 m it is still ≈ 0.8 cm/px, below the 1–2 cm/px range considered sufficient for precision surveying, so there is room to go up without losing quality.

Going up also relaxes the trigger rate. With 80 % front overlap and 70 % side overlap, the distance between photos and between lines grows with height (assuming the short side of the sensor along the route); at the current speed of 5 m/s (`WPNAV_SPEED=500`, table 3.4), the interval between triggers is that of table 3.8.

Table 3.8: *Grid* spacing and trigger interval as a function of height (25 mm, 80/70 %, 5 m/s)

H (m)	GSD (cm/px)	Between photos (m)	Between lines (m)	Interval (s)
30	0.47	3.7	8.5	0.75
40	0.63	5.0	11.3	1.00
45	0.71	5.6	12.7	1.12
50	0.78	6.2	14.1	1.25

The new `CAM1_DURATION=1.0` (table 3.5) forces a minimum interval of about 1 s between commands, so at 40 m and 5 m/s it is already at the limit, and at 30 m (0.75 s between photos) it would not be viable with these overlaps. At 45–50 m there is a small margin, and in addition fewer photos are fired in total (less data to process).

3.5.2 *Notch* and Autotune

The harmonic *notch* filter (configured from the frequency measured with FFT in *hover* and from the ESC RPM) removes the vibration noise from the control loop. With less noise, the controller does not have to spend authority correcting it and the motors run cleaner. Autotune, once the *notch* is well set, allows raising gains without reintroducing oscillation. The estimated gain is modest (5–10 %), but it costs nothing.

3.5.3 6S2P Li-ion battery

A 6S2P *pack* of 21700 cells of 4200 mAh (Molicel P42A) has ≈ 8.4 Ah at ≈ 21.6 V nominal, about 181 Wh, compared with the ≈ 89 Wh of the current 6S 4000 mAh LiPo: more than double the energy. Since the *pack* also weighs more, the net flight-time gain is estimated at ×1.7 and not ×2.

The choice of cell is critical for current. According to most distributors (figures that must be checked against the manufacturer's *datasheet* before buying), the P42A (and the P45B) supports 45 A continuous per cell, while the Samsung 50E only 9.8 A. With a *hover* of ∼17 A shared between two cells in parallel, each cell carries ∼8.5 A: the 50E would already be at 87 % of its limit just hovering, with no margin for manoeuvres or for a large *sag*, while the P42A works at 19 %.

3.5.4 Large propellers and low-KV motors

This is the dominant factor: the efficiency of a multirotor depends on the disc loading (thrust per unit of swept area), so larger propellers directly reduce the power needed in *hover*. The current 500 mm-between-axes *frame* already admits propellers of up to 13 inches, but the 1100 KV motors are not compatible with them: the change has two levels (8-inch propellers with the current motors, or 12–13 inches with 400–500 KV motors). The derivation, the KV–propeller compatibilities and the measurement protocol are developed in section 3.6. The electronics (FC, the two F9Ps, the radio and the camera) stay unchanged.

3.6 Propulsion group optimisation

3.6.1 Starting point and initial constraint

The initial propulsion configuration of the prototype was:

Table 3.9: Initial propulsion configuration

Element	Specification
Frame	Quad X, 500 mm between axes
Motors	Pilotix 2808, 1100 KV
Propellers	HQProp 7 × 4 × 3 (three-blade)
Battery	LiPo 6S 4000 mAh
AUW	2.2 kg (including the photographic payload)

The choice of the 7-inch propellers **did not respond to a design criterion but to availability**: they were propellers from a previous build and were reused to put the prototype in the air at no additional cost. This decision was reasonable in the commissioning phase – what had to be validated then was the RTK architecture, the *moving baseline* and the camera triggering chain, not efficiency – but it remained as technical debt once the system was operational. The later review of the propulsion group is the subject of this section.

3.6.2 Diagnosis

The analysis of the flight logs and the *frame* geometry revealed two inconsistencies.

The *frame* is underused. In an X configuration, the distance between adjacent motors is the diagonal divided by $\sqrt{2}$:

$$d_{\text{adjacent}} = \frac{500}{\sqrt{2}} \approx 354 \text{ mm}$$

This allows propellers of up to ∼330 mm (13 inches) with no overlap between discs. With 7-inch propellers (178 mm), **only 30 % of the available disc area** was being used.

The thrust-to-weight ratio is excessive for a mapping platform. The `MOT_THST_HOVER` parameter, learned in flight, is 0.23: *hover* is held with a low fraction of the available thrust. The resulting thrust-to-weight ratio is ∼3.1:1 when 2:1 is enough for a mapping platform. A propulsion group oversized in thrust and undersized in disc area works far from its point of maximum efficiency.

Both symptoms point to the same thing: **the propeller diameter is the limiting factor**.

3.6.3 Theoretical basis

The induced power that a rotor needs to sustain a thrust T in hover is given by momentum theory:

$$P_i = T\sqrt{\frac{T}{2\rho A}}, \qquad A = \frac{\pi D^2}{4}$$

where ρ is the air density and A the disc area. Since $P_i \propto 1/\sqrt{A} \propto 1/D$, **increasing the diameter directly reduces the power needed**. The relevant quantity is the disc loading T/A: the lower, the more efficient the rotor.

Applied to the case ($T = 550$ g per motor, $\rho = 1.225$ kg/m^3):

Table 3.10: Ideal induced power in *hover* as a function of propeller diameter

Propeller	Disc area	Disc loading	P_i per motor	P_i total
7 inches	248 cm^2	22.2 kg/m^2	50.8 W	203 W
8 inches	**324 cm^2**	**17.0 kg/m^2**	**44.5 W**	**178 W**
13 inches	857 cm^2	6.4 kg/m^2	27.4 W	109 W

The step from 7 to 8 inches reduces the ideal induced power by **12.4 %**; the step to 13 inches, by **46 %**.

These values are the theoretical limit: the real power includes the profile drag of the blades (typical figure of merit 0.65–0.75) and the efficiency of the motor-ESC assembly. In practice the total gain approaches that of the induced component, which is dominant in hover.

3.6.4 Motor–propeller compatibility constraint

The disc area cannot be increased freely: the KV of the motor must correspond to the diameter of the propeller. A motor with too high a KV for its propeller works at high torque and few revolutions, far from its point of maximum efficiency, and the aerodynamic gain is lost in electrical and thermal losses.

Table 3.11: Suitable KV as a function of propeller diameter (6S voltage)

Propeller	Suitable KV	Current motor (1100 KV)
7 inches	1200–1400	slightly low
8 inches	**950–1100**	**optimal match**
9 inches	800–900	∼25 % above
12–13 inches	400–500	incompatible

This constraint explains a counter-intuitive result: **9-inch propellers would perform worse than 8-inch ones** despite having more disc area, because the KV mismatch penalises the motor more than the disc contributes. The 8-inch propellers are therefore the optimum attainable **without replacing the motors**.

3.6.5 Propeller selection

Once the diameter was fixed, the available options were evaluated.

Number of blades. At equal diameter and thrust, a two-blade propeller is aerodynamically more efficient than a three-blade one. However, the 8-inch offer for multirotors is mostly three-blade, and the available two-blade ones (APC 8×4.5MR and MRP) have a **speed limit of ∼13,100 min^{-1}**. With 1100 KV on 6S (24,400 min^{-1} unloaded) this limit could be exceeded in full-throttle manoeuvres – gusts, recoveries, *failsafe* climbs –, with a risk of blade breakage. With a photographic payload underneath, the reliability criterion prevails over the marginal efficiency gain.

Decision: the three-blade configuration is kept and a **glass-fibre-reinforced nylon** propeller is adopted, sized for the speeds usual in high-speed multirotors.

Table 3.12: 8-inch candidate propellers

Candidate	Pitch	Mass	Notes
HQProp LR8X4.2X3	4.2 in.	–	*Long Range* range, designed for efficient cruise with a payload
HQProp MacroQuad 8×4×3	4.0 in.	12.8 g	Bore 15.7 mm, hole 9.5/5 mm (rings not included)
Gemfan 8040-3	4.0 in.	–	Equivalent, wide availability in Europe

The **HQProp LR8X4.2X3** is chosen. The pitch/diameter ratio (0.53) is lower than that of the initial configuration (0.57), which favours performance in hover and slow cruise, which is the working regime of a mapping mission.

3.6.6 Checks before mounting

- **Shaft diameter:** 5 mm, or adapter rings.
- **Camera field of view:** going from 7 to 8 inches adds 12.7 mm of radius per propeller. It must be confirmed that the blades do not enter the frame with the 25 mm lens in a nadir mount.
- **Ground clearance** with the current landing gear.
- **ESC current margin:** the current per motor decreases as the diameter increases, so no review of the sizing is needed.

3.6.7 First test with 8-inch propellers (20 October)

The first test with the 8-inch propellers ended in two aborted arms. In both, the pattern was the same: Loiter mode, take-off ramp with motor output of 43–55 %, 25–31 A, voltage above 23.85 V, and the *log* ends between 0.1 and 0.5 s later, with no *failsafe* or error recorded.

The controller stays calm and the four motors are symmetric (within ±2 %). The vibration, however, is much higher than in the previous flights: 15–35 m/s^2 in the first arm and 32–92 m/s^2 (with *clipping*) in the second, against a median of 3–5 m/s^2 and a maximum of ∼16 m/s^2 in the previous flights. The raw IMU shows a tonal vibration – narrow peaks at the rotation frequency, which grow with acceleration (127–134 Hz → 159–173 Hz) – and not a broadband one, which points to propeller imbalance (1× component) rather than rubbing. The parameters and filters were not touched for the propeller change. The exact cause of the *log* cut has not been determined; the working hypothesis is propeller imbalance, which must be ruled out by balancing them before repeating the test.

3.6.8 Control retuning

The change of diameter modifies the rotational inertia and the control authority, so the angular-rate loop gains stop being valid. After mounting it is necessary to:

1. **Redo the harmonic *notch* filter.** The fundamental frequency of the motors drops as the diameter increases. Procedure: `INS_RAW_LOG_OPT = 9`, a 60–90 s flight with varied manoeuvres, analysis with ArduPilot's *Filter Review* tool.
2. **Reduce the P and D gains** of the *roll* and *pitch* axes as a conservative starting point, and run **Autotune** axis by axis once the flight is verified to be stable.

3.6.9 Measurement protocol

To quantify the real gain, a controlled comparison protocol is established. **Prerequisite: calibration of the current and voltage sensor** (`BATT_AMP_PERVLT`, `BATT_VOLT_MULT`), without which the energy measurements are not valid. This requirement is already met (table 3.4).

Identical conditions between tests: same grid, height, cruise speed, payload, battery mass and wind conditions (< 3 m/s).

Variables to record in each test:

Table 3.13: Variables to record per test

Variable	Source
Energy consumed (mAh)	`BAT.CurrTot` in the log
Armed time (s)	`ARMED` / `DISARMED` marks
Average consumption (A)	`BAT.Curr`
Average power (W)	`BAT.Curr` × `BAT.Volt`
Efficiency (g/W)	AUW ÷ average power
`MOT_THST_HOVER`	parameter learned in flight
Area covered (ha)	mission planning

Derived metrics: mAh per minute, **hectares per battery** (the relevant quantity for the operator) and g/W (the relevant quantity for engineering).

Table 3.14: Propulsion group comparison: initial configuration (measured in flight test 6, flight 2) and 8-inch propellers (pending)

Metric	$7 \times 4 \times 3$ (initial)	$8 \times 4.2 \times 3$	Change
Energy until *failsafe* (mAh)	2,012	pending	
Time until *failsafe* (min)	10.6	pending	
Average consumption (A)	22	pending	
Average power (W)	≈ 500	pending	
Efficiency (g/W)	≈ 4.4	pending	
`MOT_THST_HOVER`	0.23	pending	
Hectares per battery at 50 m	3.1	pending	

Prior estimate: +12–18 *% endurance.*

The values in the initial column come from flight 2 of flight test 6 (section 4.3), measured with the current sensor already calibrated. The power is the product of the average current and an average voltage of ∼22.5 V. Both flights of test 6 end by low-battery *failsafe* when only 47–50 % of the nominal capacity has been used, because of the voltage drop under load (section 4.3): the usable endurance is limited by voltage, not by energy.

3.6.10 Line of continuity

The 8-inch configuration exhausts the margin available with the current motors. Making full use of the 500 mm *frame* requires replacing the motor group:

Table 3.15: Propulsion group configurations and estimated endurance

Configuration	Disc loading	Theoretical P_i	Estimated endurance
7 inches (initial)	22.2 kg/m^2	203 W	10–11 min
8 inches	17.0 kg/m^2	178 W	12–13 min
12–13 inches, 450–500 KV motors	**6.4 kg/m^2**	**109 W**	**18–22 min**

Candidates: Sunnysky X3108 480 KV or T-Motor MN3110 470 KV with 12 × 4 or 13 × 4.4 propellers. The resulting thrust-to-weight ratio (∼2.6:1) remains adequate for mapping operation. The arm mounting pattern must be verified: 28xx-series motors usually mount with a 16 × 16 mm pattern and 31xx with 19 × 19 or 25 × 25 mm.

Combined with a **6S2P Li-ion** *pack* of high-discharge cells (Molicel P42A or P45B, 206 Wh/kg versus the 143 Wh/kg of the current LiPo; section 3.5.3), the projected endurance exceeds 30 minutes.

3.6.11 Conclusion

The initial 7-inch configuration was a decision to make use of material at hand, adequate for the commissioning phase but suboptimal once the system became operational. The later analysis identifies the propeller diameter as the limiting factor of efficiency and establishes two improvements of very different cost:

- **Change to 8-inch propellers** (∼€15): uses the margin that the current motors allow.
- **Change of motor group to 12–13 inches** (∼€140): uses the existing *frame* and almost doubles the endurance.

Neither requires modifying the RTK architecture, the photographic chain or the flight software, which remain intact.

3.7 Status and pending work

3.7.1 Resolved during the campaign

- Stable control and *moving baseline* with RTK Fixed at 100 % on both receivers. Vibration, measured in the log (`VIBE` message) with the motors running, has a median of 2–5 m/s^2 and a maximum of 10.8 m/s^2 in flight test 5 and 18.7 m/s^2 in 6, with no *clipping* (table 3.6).
- Current and voltage sensor calibrated (`BATT_AMP_PERVLT`, `BATT_VOLT_MULT`); the current sensor read below the real value, which hid the real consumption until the battery collapse of 4 September.
- Camera trigger chain operational: IS-T trigger, hot-shoe *feedback* on pin 55 (110 of 110 exposures confirmed in flight 2 of test 6) and *grid* planned with the real 25 mm focal length.
- Lateral *baseline* with GPS1 on the left (`GPS1_POS_Y=`−0.19), with valid GNSS *yaw*.

3.7.2 Pending: tests and improvements

- Harmonic *notch*: measure the frequency with FFT in *hover* and configure it in mode 3 (ESC RPM); the *IMU batch sampler* is already active (`INS_LOG_BAT_MASK=1`). Verify `SERVO_BLH_POLES=14` against the real RPM, because the *notch* depends on it. Autotune afterwards.
- 6S2P Li-ion battery and, if more endurance is needed, change of motor group (section 3.5).

- Ground control and check points, measured with the same base and session as the flight, to state a precision (section 2.9).
- Compute by PPP the offset of the first session with a base log (the service already saves the stream and the *survey-in* position of every session, section 2.8) and check that the conversion to RINEX works (section 2.11.1).
- Repeat the field flight (flight test 6) with the sun more than 10–15° above the horizon (section 4.3).
- Precision test plan T1–T6 (table 2.8), with the GNSS *heading* and the accuracy of the RTK baseline; application of PPK (section 2.10).

Part III

Photogrammetric results

Chapter 4

Photogrammetry test flights

This chapter presents the two photogrammetry flights made with the complete system once the control and the GPS were stabilised (chapter 3): flight test 5, at 30 m over a park, and flight test 6, at 50 m over a field. Both ran the whole chain – planning, automatic flight with RTK, geotagging, processing and export – and serve to quantify its behaviour, in particular the shutter delay (section 4.4).

4.1 Processing chain

4.1.1 Capture

The mission is a *grid* in Auto mode at 5 m/s with distance-based triggering. The MAP2 fires the camera by PWM and the SYNC2, on the hot shoe, returns the real instant of the exposure to the controller, which records it as a `CAM` message in the *log* (the trigger command is recorded as `TRIG`). The position of each message is that of the EKF3, with RTK Fixed and GNSS *yaw*.

4.1.2 Photograph post-processing

The photographs from the camera cannot be processed as they come out:

- **Orientation.** The EXIF orientation tag of the Sony α6000 depends on the heading (the camera points at nadir), while the pixels are always 6000 × 4000. It is normalised to *Orientation*=1 in a copy: only the tag changes, the pixels are not touched.
- **Optics.** The manual lens does not record focal length, model or aperture: they are rewritten (TTArtisan 25 mm, equivalent to 38 mm, f/5.6).
- **Position.** Each photograph is paired with its *log* message (`CAM` or `TRIG`), the GPS tags are written into it and a geotag file is generated with the per-photo precision: 0.5 m on straight flight, 1.5 m in turns and 0.3 m vertically.
- **Clock.** The camera clock is off by about 9 h with respect to UTC: pairing is done by order (flight 5) or by instant (flight 6), not by the EXIF time.

The scripts that perform these steps, the flight *logs*, the corrected photographs, the geotag files and the results of each processing run are archived with SHA-256 checksums, so that each result can be regenerated from the source data.

4.1.3 Reconstruction

Reconstruction is done with ODX 3.8.3 (section 2.10). Orthophoto, DSM, point cloud and textured models are generated in EPSG:32631.

4.2 Flight test 5 – park (8 October 2026)

4.2.1 Equipment and flight

Table 4.1: Equipment and parameters of flight test 5

Item	Value
Controller	ArduCopter 4.6.3, Kakute H7, RC CRSF, ESC DShot300
Camera / optics	Sony α6000 + TTArtisan 25 mm f/2, manual focus and aperture; 1/1600 s, f/5.6
Navigation	RTK Fixed for 100 % of the flight on both receivers (GPS[0] and GPS[1]), with corrections from the own RTK base
Grid	8 passes 6.5 m apart, 7.1 m between photos (distance triggering), ∼30 m above the take-off point, ∼5 m/s
Photographs	118, between 12:26 and 12:30 (local time)
Computed overlap	75–77 % front, 65–69 % side; ∼11 images per point
Camera with respect to the drone	Rotated 90° (long side along the flight direction)
Hot-shoe *feedback*	Not available in this flight (inverted wiring, corrected and verified afterwards on the bench)

4.2.2 Results

Two processing runs were made with the same 118 photographs: the first with the GPS positions as they come out of the *log* and the second with the positions corrected for the shutter delay (section 4.4). The model geometry is the same; the georeferencing changes.

Table 4.2: Processing results of flight test 5

	Without correction	With delay correction
Photos reconstructed	118 of 118	118 of 118
Reprojection error	0.90 px	0.90 px
Sparse model points	173,084	173,401
Effective focal length (calibrated)	25.72 mm	25.74 mm
Mean GSD	0.451 cm/px	0.452 cm/px
Reconstructed area	6,633 m^2	6,714 m^2
Dense point cloud	38,547,923	38,727,582
Model–GPS difference, E / N / vert. (std. dev.)	1.30 / 0.37 / 0.12 m	**0.40 / 0.14 / 0.11 m**
CE90 / LE90 model–GPS	1.80 / 0.19 m	**0.59 / 0.18 m**
Processing time	43 min 24 s	41 min 34 s

The model–GPS errors compare the model with the photo positions: they measure internal coherence, **not absolute precision**, which requires independent check points (section 2.9). The delay correction reduces the horizontal difference from the order of 1.3 m to ∼0.4 m, and the georeferenced camera positions move at most 0.31 m between the two runs, with a model rotation of 0.2°.



Figure 4.1: Orthophoto of flight test 5 (1 cm/px, EPSG:32631), processed with positions corrected for the shutter delay.

The orthophoto (14,084 × 8,918 px at 1 cm/px) is coherent: no seams or bowl-shaped deformation, with consistent paths and roundabouts and sharp at 1:1 scale (concrete joints, individual leaves). The DSM (7,041 × 4,458 px at 2 cm/px) is clean in the interior (flat buildings, trees as relief); the first and last pass and the edges show streaks because of few views, so it is advisable to crop the DSM to the polygon of interest or to extend the flight plan one *footprint* beyond the area.

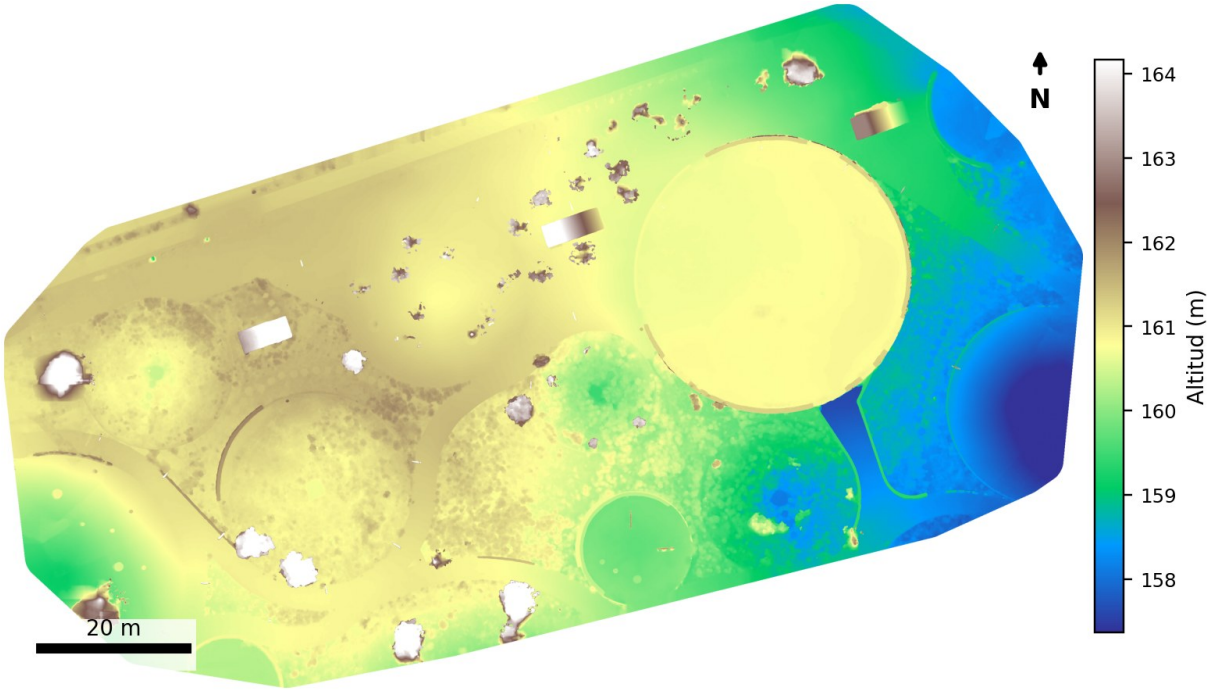


Figure 4.2: Digital surface model of flight test 5 (2 cm/px). The white gaps are treetops above the palette range.

4.3 Flight test 6 – field (13 October 2026)

4.3.1 Equipment and flights

Flight test 6 consists of two flights at ∼50 m (area of ∼180 × 150 m) with the same drone and camera as 5 and with the battery parameters already calibrated. The hot-shoe *feedback* (`CAM`) and the planning at 50 m were validated. They were made at dawn, with the sun between −3° and +1° above the horizon, so the photographs came out very dark.

Table 4.3: Log of the two flights of flight test 6

	Flight 1 (07:46)	Flight 2 (07:57)
mAh consumed (`CurrTot` / integral)	2,326 / 2,306	2,434 / 2,426
Motors running	5.5 min	11.3 min (two arms)
Voltage initial / minimum / final	24.65 / 21.25 / 21.95 V	24.73 / 21.13 / 21.95 V
Current average / maximum	25 / 56 A	22 / 58 A
Low-battery *failsafe*	268 s, 21.52 V, 1,887 mAh	635 s, 21.48 V, 2,012 mAh
Triggers `TRIG` / `CAM` / photos	94 / 93 / 94	110 / 110 / 110

- **Battery.** Both flights end by low-battery *failsafe* when only 47–50 % of the 4,000 mAh has been used: the instantaneous voltage drops below 21.6 V under load (`BATT_FS_VOLTSRC=0`). The mAh must be compared with what the charger returns.
- **Exposure.** 1/1600 s, f/5.6, ISO 1600 (ceiling of the automatic ISO): the photographs are about 6 *stops* underexposed, and the scene light is 8.5–11.9 *stops* below that of flight 5 (sun at 40°). The maximum margin of the camera is 5.7 *stops*. It is not a fault: it is a light limitation.
- **Feedback.** With `TRIG`, `CAM` and photographs in a 1:1 ratio (110 of 110 in flight 2), pairing is done by instant. There is one manual photo without `TRIG` and one `TRIG` without a photo.

Pending. Repeat this flight with the sun more than 10–15° high, which is the condition that makes it usable (the current batch only serves as a test).

4.3.2 Processing of flight 2

Flight 1 is not usable (the digital gain ×551 does not recover signal: 2 matches between consecutive photos, the same as the noise). From flight 2 the 110 photographs were processed with a linear gain of ×2 to ×97 (median ×49); **the geometry is valid, but the colours and radiometric values are not** (blue-green tint, high noise), so this batch is not suitable for colour or vegetation indices.

Table 4.4: Processing results of flight 2 of flight test 6

Item	Value
Photos reconstructed	110 of 110 (1 component)
Sparse model points / observations	101,094 / 354,361
Reprojection error	1.82 px
Mean GSD / reconstructed area	0.743 cm/px / 30,626 m^2
Dense point cloud	27,207,845 points
Camera calibration	focal length 25.79 mm
Model–GPS difference (E / N / vert.)	0.30 / 0.32 / 0.30 m; CE90 0.57 m, LE90 0.27 m
Orthophoto / DSM resolution	3 cm/px / 5 cm/px
Processing time	29 min 42 s

The quality is lower than in flight 5 (reprojection of 1.82 px versus 0.90 px): this is consistent with the noise of the images, which makes a larger working resolution (3 cm/px) preferable. The orthophoto (7,799 × 7,071 px) shows the field, the paths, the bend of the path and a white vehicle recognisably, with a blue-green tint and noise visible and an irregular western edge. The DSM (4,679 × 4,242 px) is detailed in the interior (furrows, ruts) and has streaks and an interpolation grid at the western and north-western edge.



Figure 4.3: Orthophoto of flight 2 of test 6 (3 cm/px). The images were digitally enhanced: only the geometry is valid, not the colour.

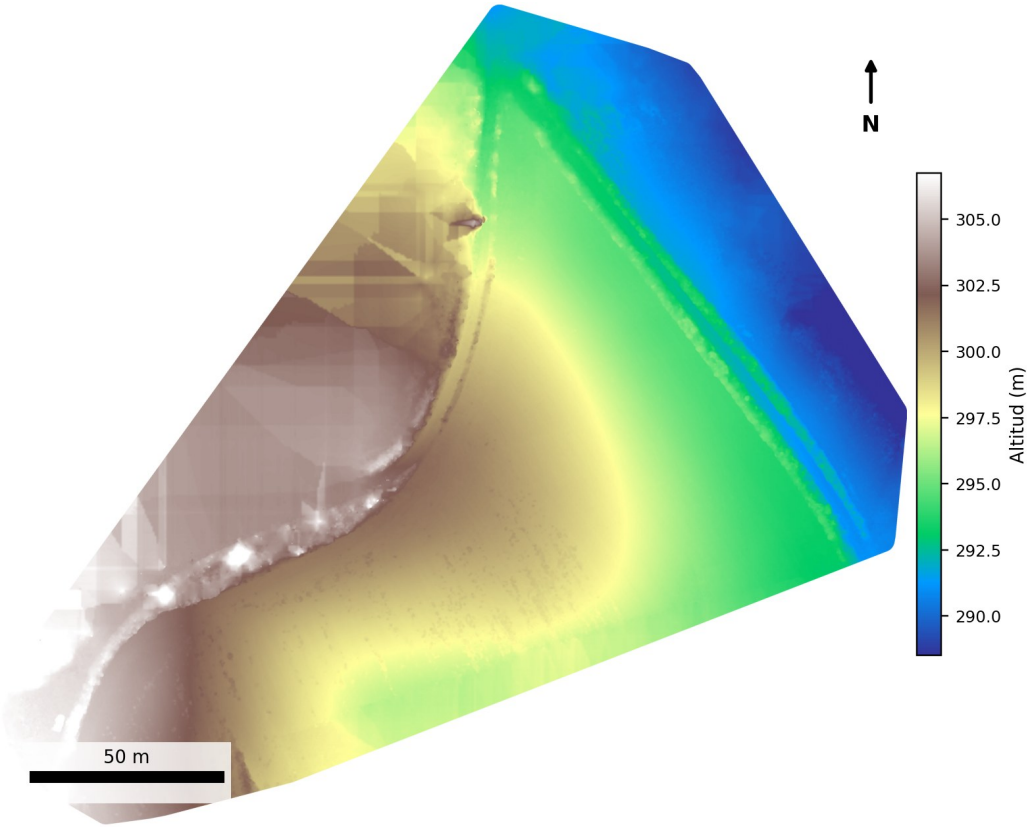


Figure 4.4: Digital surface model of flight 2 of test 6 (5 cm/px).

4.4 Shutter delay

The trigger command of the controller and the real exposure do not coincide: between them lies the delay of the MAP2, of the camera and, possibly, of the GPS/EKF position. Since the drone moves at ∼5 m/s, a delay of 0.27 s equals ∼1.3 m of displacement, far above the centimetre-level precision sought.

4.4.1 Method

For each photo, the GPS position from the *log* is compared with the camera position reconstructed by the photogrammetric model. If the delay is τ, the GPS − model residual is $-v\tau$: it has the opposite sign on the outbound and return passes and a global offset close to zero, which distinguishes it from a systematic georeferencing error.

4.4.2 Results

Table 4.5: Estimates of the shutter delay

Measurement	Delay	Notes
Flight 5, `TRIG` positions	$\tau = 0.268 \pm 0.009$ s	ENE 0.277 ± 0.007; WSW 0.267 ± 0.014
Flight 5 with corrected positions	residual 0.004 ± 0.007 s	Horizontal residual median 1.38 m → 0.35–0.49 m
Flight 6, `TRIG`→`CAM`	median 0.34 s (0.11–0.40)	Measured directly in the *log*
Flight 6, `CAM` positions	residual -0.056 ± 0.011 s	The `CAM` position is ∼56 ms ahead of the exposure
Flight 6, `TRIG`→exposure	$0.34 - 0.056 \approx 0.28$ s	Agrees with flight 5 by an independent route
Bench, command→shutter	0.12–0.19 s	Does not explain all the delay in flight

Two independent methods – the photogrammetric residual of flight 5 and the combination of `TRIG`, `CAM` and residual of flight 6 – agree on 0.27–0.28 s. It is an *effective* delay: it adds up that of the shutter and, possibly, that of the GPS/EKF position, and a fixed forward offset of the antenna with respect to the camera is confounded with it (if left as an unknown, the delay comes out at 0.34 ± 0.03 s and the offset at 0.32 ± 0.16 m, and they cannot be separated). The bench delay does not explain the whole in-flight value; it cannot be separated without *feedback*.

4.4.3 Consequences for geotagging

With *feedback* (`CAM`), the position is that of the real exposure and only a residual delay of −0.056 s remains (∼0.3 m at 5 m/s), which can be compensated. Without *feedback* the position must be shifted by $v \cdot \tau$ with $\tau \approx 0.27$ s and wider per-photo precisions. This is the reason why the hot-shoe *feedback* is mandatory for centimetre-level precision (section 1.3).

4.5 Balance of the test flights

What has worked.

- The complete chain, from planning to the DSM, with RTK Fixed throughout the flight on both receivers and valid GNSS *yaw*.
- A geometry of photogrammetric quality: reprojection of 0.90 px, 118 of 118 photos reconstructed and a GSD (0.451 cm/px) consistent with the flight height and the calibrated focal length (section 2.7).
- The hot-shoe *feedback* (110 of 110 in flight 6) and the independent quantification of the shutter delay.

Limitations.

- There are no control or check points: an absolute precision cannot be stated. The model–GPS figures are internal coherence.
- The absolute position depends on the RTK base, which in these flights ran a *survey-in* in every session with the antenna at 122 cm (section 1.5.2). The resulting position was not recorded, and the absolute error is of the order of metres, common to the whole flight.
- For flight 5 the weather conditions, the in-flight mass and the battery used are not available: the *log* and the photographs are the only record.

- Light limits the operation: flights must be made with the sun more than 10–15° high (or, with little light, 1/800 s, f/4 and ISO 3200 at the limit).
- Endurance is short: the low-battery *failsafes* act at half consumption because of the voltage drop under load (section 3.5).
- The DSM shows streaks at the periphery: the flight plan must be extended one *footprint* beyond the area of interest.

Next steps. Repeat the field flight with the sun high, measure ground control and check points to establish absolute precision, re-estimate the delay with *feedback* and improve endurance (section 3.6). The complete list of pending tasks is in section 3.7.

Pending. The absolute precision (horizontal and vertical) of the final product remains unmeasured: all the errors in this chapter are internal coherence.

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